



High Score, Low Scrutiny: A Two-Gate Quasi-Experiment on Whether Displaying a Boom Gate's Autonomous-Approval Score Changes Which Vehicles Security Actually Checks

Marek Solheim, Anouk Mensah

School of Emergent Priorities, Slop University

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Abstract. Slop University's Contactless Campus Access programme fits each boom-gated car park entrance with automatic number-plate recognition and, at one gate, a console screen showing the read's Autonomous Approval Confidence (AAC) score in real time; a second, otherwise identical gate logs the same score without displaying it. This paper reports a thirteen-week reconciliation of 6,214 vehicle-events across both gates against console-linked staff intervention and an independent desk audit of a 15% follow-up sample. Console-linked intervention is statistically indistinguishable between gates overall (35.3% visible against 37.7% hidden; n.s.), but diverges sharply within the highest confidence band, where displaying the score all but empties it of staff scrutiny (8.7% visible against 26.8% hidden). The desk audit finds the AAC score itself a weak predictor of confirmed validity across the same bands (71.4-75.9%, $r \approx 0.11$, n.s.), and a staff-rotation ablation shows the high-band scrutiny drop holds regardless of which cohort is rostered to the visible gate. Displaying an automation's own confidence signal, we suggest, recalibrates where staff choose to look rather than how accurate that choice turns out to be — a redistribution the programme's throughput figures have so far reported only as a reduction in manual handling.

Introduction

Slop University's Contactless Campus Access programme was commissioned to reduce the staff labour of admitting vehicles at the University's boom-gated car parks: automatic number-plate recognition reads each plate against the visitor and permit registers and raises the boom automatically once its Autonomous Approval Confidence (AAC) score for that read clears an internal threshold. Officers retain a console override at every gate regardless of the score, and the programme's own throughput reporting counts a falling rate of console-linked intervention as evidence the automation is doing more of the admitting work.

A falling intervention rate is consistent with at least two explanations the throughput figure alone cannot separate. The first is that the au-

tomation genuinely admits a safe vehicle more often than it once needed a human's help to. The second is that staff attention has simply moved — toward the reads the system itself flags as uncertain, and away from the reads it flags as confident, whether or not that confidence is earned. Literatures on automation reliance and trust calibration document exactly this second pattern in other supervisory-control settings: operators shown a system's own confidence or state recalibrate how closely they watch it, and the recalibration is not guaranteed to track how reliable the system actually is [1], [2], [3]. Whether Slop University's own gates show the same pattern, and whether the confidence signal they recalibrate against is itself any good, has not previously been checked against an audit independent of the gate's own log.

This paper reports a thirteen-week field audit reconciling console-linked staff intervention at two otherwise matched gates — one where the AAC score is shown to staff, one where it is logged but withheld — against a desk audit of confirmed vehicle validity drawn independently of what happened at the boom. Its contributions:

1. We report the first reconciliation, at this institution, of a boom gate's displayed confidence score against staff intervention and an independent validity check, rather than against the gate's own throughput count.
2. We find console-linked intervention converges between the two gates overall but diverges sharply in the highest confidence band, where display of the score all but empties it of scrutiny.
3. We show, via a staff-rotation ablation, that the divergence is not attributable to which officers are rostered to the visible gate.

Related work

Supervisory control research has long distinguished an automated system's actual reliability from the reliance an operator extends to it, the two governed by partly independent processes [1], with misuse, disuse, and abuse of automation traced to exactly this gap [4]. Displaying a system's own state is one condition shown to shift that reliance: monitoring of a trusted automated stage is shown to decline, a pattern characterised as automation complacency [2], and mode-awareness failures in supervisory control are shown to arise from a gap between what a system is doing and what its operator believes it is doing [3].

The asymmetry runs both directions: people are shown to abandon an algorithm's advice after witnessing a single visible error even where the algorithm remains more accurate than the human alternative [5], and, in other decision settings, to defer to an algorithm's judgement over an equivalent human one [6]. Automated recommendations are shown more broadly to draw less scrutiny than an equivalent human recommendation would, an effect termed automation bias [7], and lessons drawn across decades of human-automation research caution that a taxonomy of automation levels does not, by itself, predict how an operator will actually supervise a given system [8], [9].

Where responsibility for an admitted vehicle sits once a human retains only an override is characterised as a moral crumple zone: accountability concentrates on whichever party keeps the final manual action, independent of how much judgement that party was still exercising [10], a displacement Bainbridge's earlier account of automation's ironies traces to the same source: practised judgement pushed onto ever-rarer manual moments [11]. Meaningful human control is argued to require more than a retained override button, if the conditions under which that button gets pressed are themselves shaped by the automation's own displayed output [12]. The plate-reading layer beneath the confidence score is itself an active recognition research area independent of any display question [13], [14], [15].

Prior work at this institution has already found staff overriding an automated arrival forecast for a reason the forecast had itself already logged [16], and an automated car-park priority score whose rank tracked how confidently a form was filled in rather than the commuting need it was meant to represent [17]; this paper asks the same question of a gate rather than a schedule or a form.

Method

Two gates

The audit ran across the Eastern Approach and Service Yard boom gates, matched by Facilities for comparable vehicle volume and identical hardware and AAC scoring logic. Only the Eastern Approach gate carries a console-mounted display of each read's AAC score; the Service Yard gate logs the same score to the same database but shows officers nothing beyond the boom's own automatic or manually-triggered state.

Event logging and the intervention rule

Every plate read, its AAC score, and the boom-rise event that followed it were logged automatically across a thirteen-week teaching period (6,214 vehicle-events; Table 1). A read counts

as console-linked intervention if any officer console action — a manual buzz, a re-scan request, or a flagged hold — is logged within the twenty seconds preceding the boom’s rise:

$$\text{Intervention rate} = \frac{N_{\text{console-linked}}}{N_{\text{total}}} \times 100$$

Scores were banded into Low (0-39), Medium (40-74), and High (75-100) following the programme’s own published threshold for automatic admission at 75.

Desk audit

A random 15% of events (933 of 6,214) were re-verified independently of the gate log: each vehicle’s plate was checked against Facilities’ visitor and permit records the same afternoon by a researcher blind to which gate had logged the event, its AAC score, or whether it had drawn a console-linked intervention. This yields a validity estimate for each confidence band that owes nothing to what officers happened to do at the boom.

Staff-rotation ablation

The fourteen officers rostered across both gates were assigned to one of two rotation cohorts, alternating which cohort staffed the Eastern Approach gate week to week across the study window (cohort A: seven weeks; cohort B: six weeks). If the score-display effect reported below reflects one cohort’s particular habits rather than the display itself, the effect should separate by cohort; if it reflects the display, it should not.

Results

Gate	Low	Medium	High	Total
East-ern Ap-proach (visible)	950	1,750	402	3,102
Service Yard (hid-den)	960	1,760	392	3,112

Table 1: Audited vehicle-events by gate and AAC confidence band.

Overall console-linked intervention across the thirteen weeks was statistically indistinguishable between gates: 35.3% at the Eastern Approach against 37.7% at the Service Yard ($\chi^2(1, N = 6214) = 2.87, p = .090$), short of the study’s pre-registered $p < .05$ threshold. That aggregate similarity masks a divergence concentrated entirely in the top band (Figure 1). Low- and Medium-band intervention rates track closely between gates (Low: 58.1% visible against 58.0% hidden; Medium: 29.0% against 29.0%), but High-band intervention falls to 8.7% at the Eastern Approach against 26.8% at the Service Yard — a gap the Low and Medium bands, run at the same two gates under the same rostered officers, do not show.

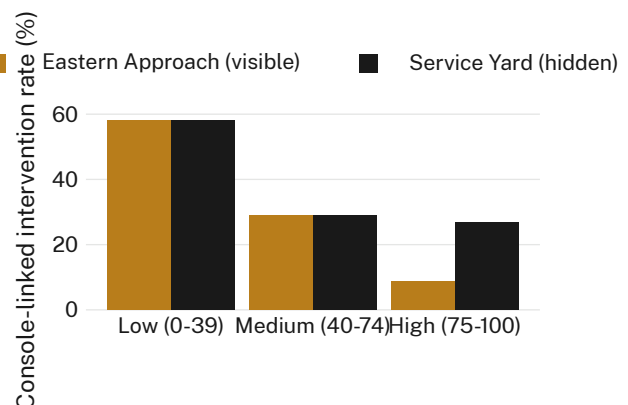


Figure 1: Console-linked intervention rate by AAC confidence band and gate. Low- and Medium-band rates track closely between gates; the High band diverges sharply only where the score is shown to staff.

The desk audit’s independent validity estimate rises only gently across the same three bands — 71.4% (Low), 73.6% (Medium), 75.9% (High) — and a linear fit across the finer-grained score

deciles behind that aggregate is weak ($r \approx 0.11$, n.s.; Figure 2). A High-band read is, on this evidence, barely more likely to belong to a vehicle the desk audit would confirm than a Low-band read is; the near-emptying of scrutiny the display produces in that same band is not obviously earned by the score's own accuracy.

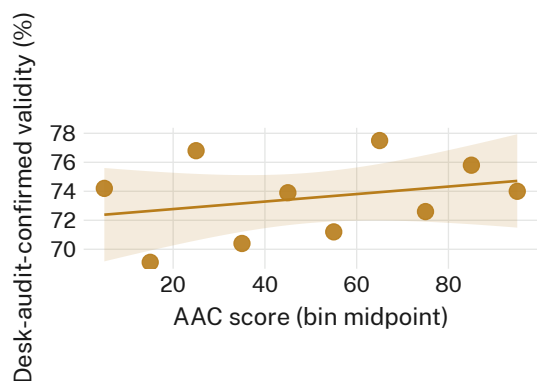


Figure 2: Desk-audit-confirmed validity rate against AAC score, in ten-point bins pooled across both gates. The fitted trend is weak across the score's full range.

The staff-rotation ablation changes almost nothing. High-band intervention at the Eastern Approach gate, split by which cohort was rostered that week, shows overlapping distributions (cohort A: median 8.9%; cohort B: median 8.7%; Figure 3)—consistent with a display effect rather than a particular team's habits, and we retain both cohorts pooled in the headline comparison above for that reason.

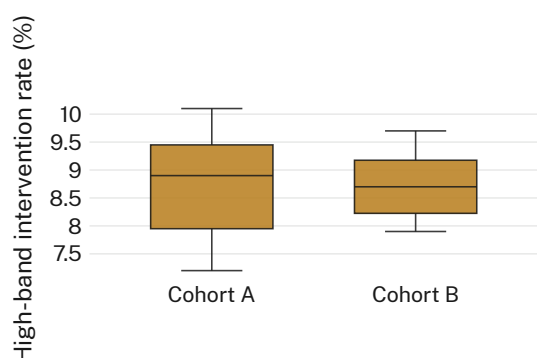


Figure 3: Weekly High-band intervention rate at the Eastern Approach gate, by staff-rotation cohort. The two cohorts' distributions overlap almost completely.

Discussion

Two independent measures agree that the Eastern Approach gate's near-empty High-band scrutiny is not explained by the High band

actually being safer. The desk audit's validity estimate rises only gently across the AAC score's range, and the ablation rules out the rostered team as an alternative explanation. What the comparison across gates does support is that showing staff the automation's own confidence signal changes where their attention goes, in a pattern consistent with reliance and complacency findings from other supervisory-control settings [1], [2] rather than with a genuine improvement in what the automation is admitting.

None of this shows the Contactless Campus Access programme's throughput figures to be false. Console-linked intervention has fallen at the Eastern Approach gate, exactly as the programme reports, and exactly as intended. What the figures do not show, and were not designed to show, is that the fall reflects the automation doing safer admitting rather than staff doing less looking at the reads the automation itself tells them not to worry about. A confidence score displayed without an accompanying accuracy readout invites the first reading; this audit's evidence favours the second.

Limitations

This audit covers two gates at one institution across a single thirteen-week period; a longer window might surface a genuine High-band safety improvement this study's desk-audit sample was not large enough to detect, though the consistency of the pattern across all fourteen rostered officers suggests the effect is not confined to a small handful of shifts. The desk audit re-verified 15% of events rather than all 6,214, a rate chosen to keep the audit itself from becoming a second intervention at the gate it was checking. Officers were not told which gate the audit concerned, which protects the comparison but leaves open that some had already formed their own view of the display's reliability before this study began.

Conclusion

Displaying an automated gate's own confidence score changed how much scrutiny the highest-confidence reads received without changing, on independent evidence, how much scrutiny they deserved. Console-linked intervention converged between a score-visible and a score-hidden gate overall and diverged sharply in exactly the band the display was meant to reas-

sure staff about, a divergence the staff-rotation ablation could not attribute to any one rostered team. The programme's own throughput count, reporting only the fall in manual handling, cannot currently distinguish this redistribution from the automation improvement it is generally read as evidence of. Future work might test whether pairing the displayed score with a periodic validity readout closes the scrutiny gap this audit finds open.

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